

766

Tests Passing

45

Test Files

14K

Lines of Code

4

Languages

March - April 2026 | v2.0 Multi-Channel Release

| Component             | Status     | %    | Notes                                 |
|-----------------------|------------|------|---------------------------------------|
| Telegram Booking Core | Production | 100% | Multi-tenant, roles, appointments     |
| Stripe Billing        | Production | 95%  | Subscriptions, webhooks, grace period |
| AI Chat (Workers AI)  | Production | 90%  | 3-model fallback, action tags         |
| Google Calendar       | Production | 95%  | OAuth, sync, watch renewal            |
| Support Tickets       | Production | 100% | Platform + tenant-local tickets       |
| i18n (RU-UA-EN-PL)    | Production | 100% | 350+ keys, 4 languages                |
| WhatsApp Channel      | Partial    | 80%  | Adapter done, needs E2E testing       |
| Instagram Channel     | Partial    | 75%  | Adapter done, needs E2E testing       |
| Admin-App (Next.js)   | Partial    | 85%  | Dashboards done, channels UI missing  |
| Multi-Channel UX      | Scaffolded | 60%  | Schema ready, UI not built            |
| E2E Tests             | Minimal    | 20%  | No Playwright/Cypress tests yet       |

## Arkitektura / Architecture

| Layer         | Technology               | Details                                         |
|---------------|--------------------------|-------------------------------------------------|
| Runtime       | Cloudflare Workers       | Edge compute, <30ms cold start, 30s timeout     |
| Database      | Cloudflare D1 (SQLite)   | Multi-tenant, 15+ tables, proper indexes        |
| Cache / State | Cloudflare KV            | User state, locks, encrypted tokens (TTL-based) |
| Admin App     | Next.js 15 + tRPC 11     | 19 routers, Drizzle ORM, Tailwind CSS 4         |
| AI            | Cloudflare Workers AI    | Llama 4 Scout -> Llama 3.1 8B fallback chain    |
| Payments      | Stripe API (fetch-based) | No SDK, webhooks with HMAC verification         |
| Channels      | Custom Adapter Layer     | Telegram + WhatsApp + Instagram adapters        |

## 2. Nedelya 1 (28 marta - 3 aprelya)

### Multi-Channel Finalization

#### Den 1-2: End-to-End Channel Testing

- Apply migration 0002\_multi\_channel.sql to production D1
- Register Meta App in Business Manager (one app for WA + IG)
- Configure META\_APP\_SECRET, META\_VERIFY\_TOKEN\_WA, META\_VERIFY\_TOKEN\_IG secrets
- Test WhatsApp inbound: text, buttons (reply + list), contacts, images
- Test Instagram inbound: text, quick\_reply, postback, image attachments
- Verify 24h message window tracking (message\_windows table)
- Fix normalization bugs found during real traffic testing

#### Den 3-4: Channel Configuration UI (Admin-App)

- Add "Channels" tab to SalonDashboard.tsx
- WhatsApp config form: phone\_number\_id, access token, verify token
- Instagram config form: page\_id, access token
- Token encryption on save (reuse encryptAndStoreToken from token-manager.js)
- Channel enable/disable toggle with active flag
- Display channel connection status + last message timestamp
- Meta Embedded Signup widget integration (OAuth flow for salons)

#### Den 5: Conversations UI

- Build /conversations page in admin-app
- List conversations across all channels (Telegram, WA, IG) - unified view
- Filter by channel type, status (open/closed), date range
- View full conversation message history with channel badges
- Wire up tRPC conversations router (already implemented server-side)

#### Deliverables / Rezultaty nedeli

| Deliverable               | Priority        | Depends On                       |
|---------------------------|-----------------|----------------------------------|
| Migration applied to prod | <b>CRITICAL</b> | wrangler d1 migrations apply     |
| WA/IG inbound tested E2E  | <b>CRITICAL</b> | Meta App registration            |
| Channel config UI         | <b>HIGH</b>     | Migration + tRPC channels router |
| Conversations page        | <b>HIGH</b>     | tRPC conversations router (done) |

## 3. Nedelya 2 (4 - 10 aprelya)

### Security & Quality Hardening

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#### Den 1-2: Security Hardening

- Hard rate limiting: reject requests after quota exceed (HTTP 429)
- Request size validation in worker.js (reject >1MB payloads)
- WhatsApp template quota: hard enforcement (block send at monthly cap)
- Input sanitization audit: all user-facing inputs validated
- Admin-app: add request signing for sensitive mutations
- Audit all token storage: verify encryption at rest, no plaintext in logs

#### Den 3-4: Test Coverage Expansion

- Install Playwright for E2E testing
- E2E: Booking flow (service -> date -> time -> master -> confirm)
- E2E: Admin panel navigation + appointment management
- E2E: Support ticket lifecycle (create -> claim -> reply -> close)
- E2E: Language switching across all 4 languages
- Integration tests: WhatsApp inbound -> normalize -> handleInbound -> response
- Integration tests: Instagram inbound pipeline
- Target: 800+ total tests

#### Den 5: Google Calendar Edge Cases

- DST transition handling (Warsaw timezone)
- Watch renewal retry logic (exponential backoff on failure)
- Configurable sync horizon per tenant (default 90 days)
- Error recovery for partial sync failures (transaction rollback)

## 4. Nedelya 3 (11 - 17 aprelya)

### Admin-App Polish & Analytics

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#### Den 1-2: Analytics Dashboard

- New /analytics page with Recharts visualizations
- Monthly revenue line chart (Stripe data)
- Daily bookings bar chart + booking heatmap (hour x day)
- Top services ranking, busiest hours, conversion rates
- Channel distribution pie chart (Telegram vs WA vs IG)

#### Den 3-4: Detail Pages

- Master Performance Dashboard: earnings, schedule, client count
- Client detail page: booking history across channels
- Communication log: unified message history per client
- Webhook event log viewer (debug incoming webhooks, replay failed)

## Den 5: UX Polish

- Mobile responsiveness audit (all admin-app pages)
- Dark/light theme toggle (persist in localStorage)
- Notification badges: new tickets, pending appointments
- Real-time updates via polling (15s interval for dashboards)

## 5. Nedelya 4 (18 - 24 aprelya)

### Production Hardening & Launch Prep

#### Den 1-2: Performance

- Caching layer: tenant config cached in KV with 5-min TTL
- Pagination for all list endpoints (appointments, users, conversations)
- Database query audit: EXPLAIN on slow queries, add missing indexes
- Batch D1 operations where possible (dbBatch for multi-insert)

#### Den 3-4: Observability

- Structured logging: JSON format with correlation IDs per request
- Health check endpoint: GET /health (DB + KV connectivity)
- Cloudflare Workers analytics dashboard setup
- Error alerting: Telegram notification to ADMIN\_CHAT\_ID on 5xx errors

#### Den 5: Pre-Launch

- Final security audit: OWASP checklist for API
- Load testing: simulate 100 concurrent tenants, 1000 bookings
- Documentation: API reference, deployment guide, architecture diagram
- Rollback plan documented

## 6. Mesyachnyj plan - May 2026

### Growth Features

| Week                  | Focus Area       | Key Deliverables                                                                  |
|-----------------------|------------------|-----------------------------------------------------------------------------------|
| Week 1<br>(May 1-7)   | AI Enhancement   | Streaming responses (SSE) + conversation memory via vector embeddings (Vectorize) |
| Week 2<br>(May 8-14)  | New Channels     | SMS adapter (Twilio) + email notification system (Resend/SES)                     |
| Week 3<br>(May 15-21) | Advanced Billing | Usage-based pricing tiers, proration on plan changes, dunning email automation    |
| Week 4<br>(May 22-28) | Platform API     | Public REST API for third-party integrations + outbound webhook delivery system   |

## 7. Priority / Priority Matrix

### CRITICAL - Nedelya 1

|                                                          |
|----------------------------------------------------------|
| Apply 0002_multi_channel.sql migration to production D1  |
| Test WhatsApp + Instagram inbound E2E with real accounts |
| Build Channel Configuration UI in SalonDashboard         |
| Build Conversations UI page                              |

### IMPORTANT - Nedeli 2-3

|                                                 |
|-------------------------------------------------|
| Hard rate limiting enforcement (HTTP 429)       |
| E2E test suite (Playwright) - target 800+ tests |
| WhatsApp template quota hard cap                |
| Analytics dashboard with Recharts               |
| Master Performance + Client detail pages        |
| Google Calendar DST + watch renewal fixes       |

### NICE TO HAVE - Nedelya 4+

|                                |
|--------------------------------|
| AI streaming responses (SSE)   |
| SMS channel adapter (Twilio)   |
| Advanced analytics + reporting |
| Public API + webhook delivery  |
| Performance caching layer      |

## 8. Metriki uspekha / Success Metrics

| Metric                     | Current    | Target (April 30)  | Status        |
|----------------------------|------------|--------------------|---------------|
| Total tests                | 766        | 800+               | On track      |
| Critical security findings | 0          | 0                  | Achieved      |
| WhatsApp channel           | Scaffolded | Live in production | In progress   |
| Instagram channel          | Scaffolded | Live in production | In progress   |
| Conversations UI           | Not built  | Complete           | Not started   |
| E2E test suite             | None       | Core flows covered | Not started   |
| p95 response time          | ~500ms     | <2s under load     | Needs testing |

| Metric     | Current | Target (April 30) | Status   |
|------------|---------|-------------------|----------|
| Uptime SLA | 99%+    | 99.9%             | On track |

### Tech Stack

**Worker:** Cloudflare Workers + D1 + KV + Workers AI | **Admin:** Next.js 15 + tRPC 11 + Drizzle + Tailwind 4 | **Payments:** Stripe | **Channels:** Telegram + WhatsApp Cloud API + Instagram Messaging API | **Calendar:** Google Calendar API (OAuth 2.0) | **Tests:** Vitest | **Deploy:** wrangler deploy + GitHub Actions